

NE8 Assignment 4: WIMS

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The code for all the exercises can be found in `aqz22.code.zip`. Inside there are folders numbered 1 to 7 corresponding to the 7 exercises. The `input` files contain the inputs for the WIMS computations. All calculations for the input parameters (such as fuel rod radii, ratio of nuclides, etc.) were done in python. This code can be found in `aqz22.inputCalculations.py`. The code for generating the plots can be found in `aqz22.outputPlots.py`.

There are several variables that were not given in the assignment. For the moderators: the density of H_2O is 0.66639 g/cm^3 and the density of D_2O is 0.73191 g/cm^3 ; these were obtained by interpolating the *Density of Water and Steam* table in the CUED Thermofluids Data Book, as well as Table 3 in IAEA: Tables of thermodynamic properties of heavy water in S.I. units. Atomic masses were obtained from the IAEA Live Chart of Nuclides.

- 1 **Perform criticality calculations for different V_m/V_f by changing the pin lattice pitch, starting from the ‘simple cell’ example. Use the specifications in Appendix A. Do this for 2 different moderator materials: a) H_2O b) D_2O . Increase the pitch to observe both over- and under-moderated lattice configurations. Plot k_{∞} of the cell as a function of V_m/V_f .**

A total of 19 WIMS simulations were run with different lattice pitches. The relation between k_{∞} and V_m/V_f for H_2O and D_2O are plotted in Figure 1.

Comment on the results.

Looking at the light water pin in Figure 1a, the reactivity peaks at around $V_m/V_f = 5$. The reference pin has $V_m/V_f = 2.0136$ so it is undermoderated. This is a good thing because it means that the moderator temperature coefficient is negative, thus making the reactor stable.

The reference pin for heavy water in Figure 1b is also undermoderated, but the peak is at a much higher value around $V_m/V_f = 300$. The reason for this is related to the much lower capture cross-section of D_2O . Moderators do two things: they improve the neutron economy by softening the neutron spectrum, and they worsen the neutron economy by absorbing neutrons. Above the V_m/V_f peak, the additional neutron capture dominates the gains from a softer spectrum, thus leading to less reactivity. Since deuterium has a very low capture cross-section, there is much more deuterium necessary before neutron capture becomes significant. Thus the V_m/V_f peak is much higher than compared to light water.

It can also be observed that the peak k_{∞} is higher for D_2O compared to H_2O . This is also because of the lower capture cross-section. D_2O has a better neutron economy which increases the reactivity.

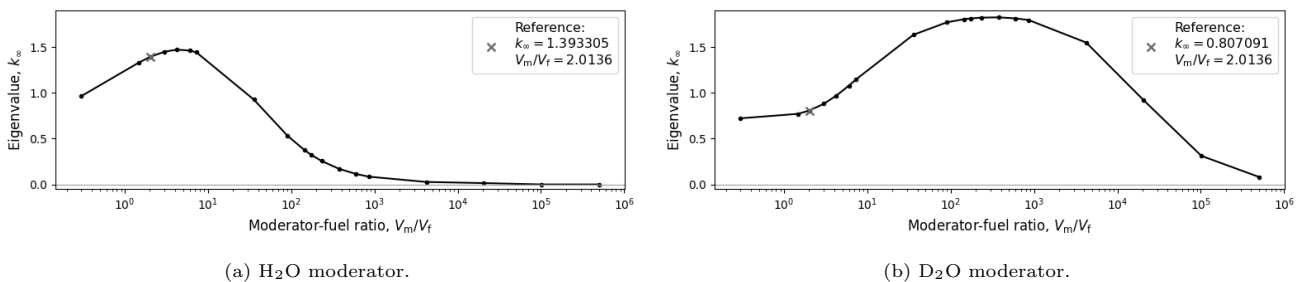


Figure 1: Linear-log graphs of k_{∞} at different ratios of fuel and moderator volumes. The two plots correspond to different moderators. Both have a temperature of 600 K and a pressure of 15.5 MPa. The gray crosses indicate the reference pin with a pitch of 1.260 cm as described in Appendix A of the assignment.

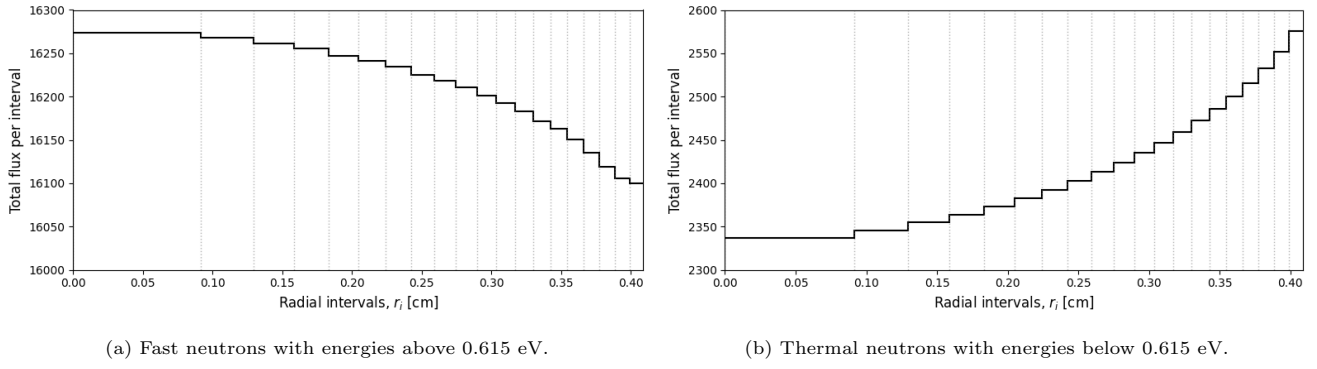


Figure 2: Radial distribution of the neutron flux in the reference pellet with radius $R = 0.4095$ cm. The plot depicts the total flux rate per radius interval. The edges of each interval is indicated by the dotted vertical lines. Each radius interval represents a surface area of 0.026341 cm^2 .

2 For standard unit cell dimensions, divide the fuel pin into a number of radial equal-volume sub-regions (rings) obtain radial distribution of fast and thermal neutron flux.

A total of 20 equal volume radial regions were found using Equation 1:

$$r_i = R\sqrt{i/10} \quad (1)$$

where R is the outer diameter of the fuel pellet, and r_i is the i -th equal-volume radius. A new copy of the fuel material was made for each radial interval.

Use 0.615 eV energy as a cut-off between the fast and thermal energy regions. Plot the radial variation of fast and thermal fluxes.

The radial variation of fast and thermal fluxes is shown in Figure 2.

Comment on the results.

Comparing the fast neutrons in Figure 2a with the slow neutrons in Figure 2b, shows that the spectrum is harder near the center of the pellet. This makes sense because the pellet is surrounded by a moderator. Neutrons are (mostly) slowed down by the H_2O moderator outside the pin. As these slower neutrons enter the pin from the outside, they will first be absorbed by the outer uranium. When a fission occurs, neutrons are born fast again. In other words: inbound thermal neutrons get absorbed as they get closer to the center, while neutrons born in the fuel are fast. The combination of these effects results in a harder spectrum near the center.

A more subtle observation is that the rate of change in flux decreases near the pellet center. This is because inbound neutrons with energies near absorption peaks get absorbed as they approach the center, thus reducing the effective absorption cross-section. This phenomenon will be explained in more detail in the next exercise.

3 Revert to your original one-region pin. Carry out a study of the effect on k_{∞} of changing the radius of the fuel whilst keeping V_m/V_f constant. Also, find and plot the one-group capture cross section of U238 as a function of pin radius.

The fuel pin radius R was varied to 15 different values between 0.001 cm and 20 cm. The resulting k_{∞} for the different radii is shown in Figure 3a. The different microscopic capture cross-sections σ_c are plotted in Figure 3b.

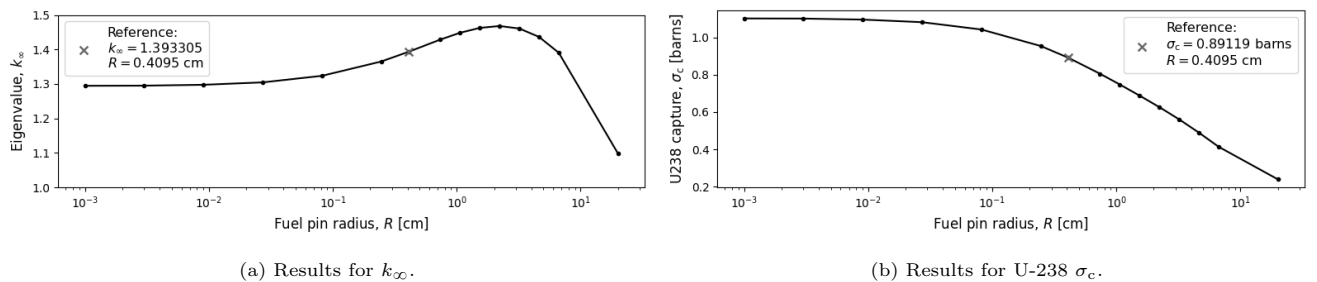


Figure 3: Linear-log graphs depicting simulation results for fuel pins by varying the pellet radius R . The gray crosses indicate the reference pin as was described in Appendix A of the assignment. The moderator-fuel ratio was kept a constant $V_m/V_f = 2.0136$, the same value as the reference pin.

Explain what you find.

Figure 3a suggests that there is a certain optimal radius around $R = 2$ cm. Before explaining the causes of this peak, it is important to analyze the neutron lifecycle. A neutron is born fast; then it moves from the fuel to the moderator where it slows down; at last it reenters the fuel and fissions. That is the ideal case. A neutron can also be captured without fission: inside the fuel by U-238, and inside the moderator by hydrogen. U-238 capture is more likely when the neutron enters the fuel while in the epithermal region. Hydrogen capture is more likely if the neutron stays in the moderator for a long time. The combination of these two effects means that there is some optimal amount of time for the neutron to stay in the moderator that minimizes the chance of capture. By increasing the radius of the pin, the scale of entire setup increases. That is to say: a larger value of R , means that neutrons are less likely to cross between the fuel and moderator boundary. Thus, R determines the average amount of time neutrons spend inside the moderator before reentering the fuel. As discussed before, there is an optimal amount of time for a neutron to stay in the moderator to minimize the chance of capture. It follows that there is an optimal value for R to maximize k_{∞} .

Figure 3b shows that the U-238 σ_c decreases for larger R . This can be explained by noting that the U-238 capture cross section is highest at the resonance peaks in the epithermal region. As a result, neutron capture tends to happen after the neutron has undergone some scatter events. Such neutrons must enter the pin from the outside, because that is where the moderator is. As they do so, they will first be absorbed by the outer U-238. This leaves the inner U-238 with fewer neutrons at the absorption peaks, because these peaks are filtered out as these neutrons pass through the outer U-238. This effect is exacerbated by increasing R , as there is now more inner U-238 that will not see neutrons at the resonance peaks. In summary, σ_c is lower for higher R because neutrons need to pass through more U-238 where they are captured at the resonance peaks. This leaves fewer neutrons at the resonance peaks, reducing the overall chance of capture.

- 4 *Perform burn-up calculations for the reference fuel unit cell. (...) It is advisable when performing burnup to take some short initial burnup steps to reach equilibrium for important products like xenon. Use IF statements so that the first step is 3 days long, the second step is 15 days, and then a further 10 to 20 steps to the end of burnup (choose an appropriate total burnup time)*

Burnup calculations were done with a first step of 3 days (consisting of 10 substeps of 0.3 days), and a second step of 15 days (consisting of 10 substeps of 1.5 days). After this initial burnup, a further 19 steps of 30 days were performed (consisting of 10 substeps of 3 days). In total, the fuel was burned for 588 days, or about 17.5 months. Typical PWR fuel lasts 12-18 months, so this was seen as an appropriate number.

Plot nuclide density as a function of burnup for the following isotopes: U238, U235, Pu239-242.

The nuclide burnup plots are shown in Figure 4. As the fuel burns up, the concentration of fissile U-235 decreases. This can be seen in Figure 4a. Figure 4b shows that the concentration of U-238 also decreases over time. This is in large part due to neutron capture. The resultant increase in plutonium can be observed in Figure 4a.

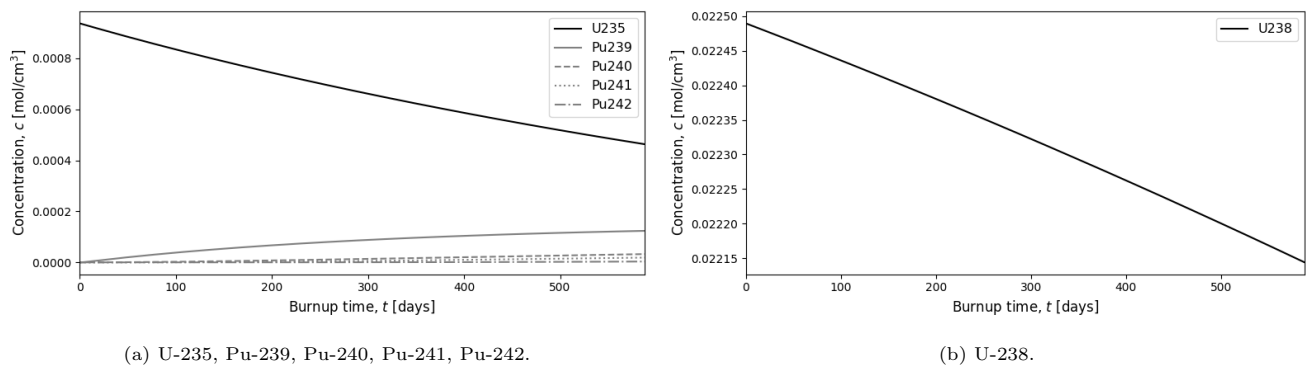


Figure 4: Graphs showing the concentration for various nuclides during fuel burnup. The black lines are the uranium isotopes. The gray lines are the plutonium isotopes.

5 *Start from the ‘assembly’ example case. You do not need to modify the geometry or material specification. First, use the Method of Characteristics in 172 energy groups. Investigate how the calculated value of k_{∞} varies with changes to the ray spacing and angle. Perform a parametric study by varying the number of angles and ray spacing until the changes in the k_{∞} results are negligibly small or no longer observable. You should thereby produce a ‘reference’ solution with a suitable number of azimuthal and polar angles and sufficiently small spacing between characteristic rays.*

There are 3 parameters that need to be chosen: the ray separation distance, the number of azimuthal angles, and the number of polar angles. They were studied in that order. Ultimately, the solution converged for: a ray separation distance of 0.005 cm, 9 polar angles (the maximum), and 100 azimuthal angles. The result is shown in the first row of Table 1.

6 *Use the collision probabilities method using 172 energy groups and compare it to the results obtained using the Method of Characteristics, both in terms of k_{∞} and calculation time.*

The results for k_{∞} , as well as the runtimes are shown in the first two rows of Table 1. It is impressive that PERSEUS has an error of less than 0.15%, while using 18000 times less runtime.

7 *Now try a hybrid of the two methods with the following stages: 1) solve the multi-cell collision probability equations, 2) obtain neutron spectra for each material, 3) generate cross sections in fewer energy groups (condensation), and finally 4) calculate the few-group fluxes by the Method of Characteristics. Here, you are required to use different number of energy groups in stage 3 (i.e. 1, 6, 11 and 22 groups) for condensation.*

The hybrid method was tested using 1, 6, 11, and 22 energy groups as suggested by the exercise. The results are shown in Table 1.

Identify the trade-offs of using different methods/sequences on the efficiency and accuracy by comparing: difference in reactivity, simulation (CPU) time

The most obvious conclusion from Table 1 is that the full Method of Characteristics is overkill for most situations. It takes the longest amount of time by a huge margin, while all other solutions are within 0.3% of the answer. The Method of Collision Probabilities is actually remarkably accurate considering its speed. For this reason, I would expect this method to be used for applications where speed is important.

For cases where accuracy is needed, the hybrid methods can yield better results. The caveat is that this is only the case when the number of energy groups is sufficiently high. I would say that the hybrid method with 22 energy groups would be a good candidate for an accurate method that is still reasonably quick.

A surprising finding is that the one-group method is more accurate than some of the multi-group methods. I hesitate to draw any conclusions from this, as this is just one test, so perhaps this is just an anomaly for this particular setup¹. However, if it is generally true that the one-group approximation is more accurate than low-resolution multi-group simulations, then this method would be excellent for providing more accuracy without too much more computation time.

Table 1: Different WIMS transport methods and their resulting k_{∞} and runtime. The “ k_{∞} error” refers to the relative difference compared to the Method of Characteristics. It was calculated using $100\% \cdot |k_{\infty}^{\text{MoC}} - k_{\infty}|/k_{\infty}^{\text{MoC}}$, where k_{∞}^{MoC} is the reference solution.

Method	k_{∞}	k_{∞} error	Runtime
Method of Characteristics (CACTUS)	1.176048	<i>Reference</i>	359.47 s
Method of Collision Probabilities (PERSEUS)	1.174293	0.1492%	0.02 s
Hybrid with 1 energy group	1.175038	0.0762%	0.53 s
Hybrid with 6 energy groups	1.173053	0.2450%	5.84 s
Hybrid with 11 energy groups	1.173532	0.2043%	11.26 s
Hybrid with 22 energy groups	1.176011	0.0065%	37.21 s

¹This is really true for the entire exercise. To truly form a conclusion you would have to test multiple different setups, including more complicated configurations.